Here are some terms to familiarize yourself with:

Random Sampling: Usually pertains to a population and is used to get an unbiased and representative sample. Random sampling can be used to support valid inferences if the sample is representative of the population.

Experimental Probability: Found by repeating an experiment and observing the outcomes.

$$P(\text{event}) = \frac{\text{number of times event occurs}}{\text{total number of trials}}$$

Example:
A coin is tossed 10 times: A head is recorded 7 times and a tail 3 times.

$$P(\text{head}) = \frac{7}{10}$$

$$P(\text{tail}) = \frac{3}{10}$$

Theoretical Probability: Expresses the likelihood that something favorable will occur.

$$P(\text{event}) = \frac{\text{number of favorable outcomes}}{\text{total number of possible outcomes}}$$

Example:
A coin is tossed.

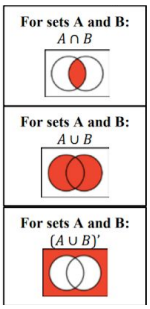
$$P(\text{head}) = \frac{1}{2}$$

$$P(\text{tail}) = \frac{1}{2}$$

Conditional Probability and the Rules for Probability

Here are some set operations to know:

Intersection: the intersection of two sets A and B is the set of elements that are common to both set A and set B. It is denoted by $A \cap B$ and is read "A intersection B"



Union: the union of two sets A and B is the set of elements, which are in A or in B, or in both. It is denoted by $A \cup B$, and is read "A union B"

Complement: the complement of the set $A \cup B$ is the set of elements that are members of the universal set U but are not in $A \cup B$. It is denoted by $(A \cup B)'$

Conditional Probability: A measure of the probability that an event occurs given that another event has already occurred. (ex. the probability of someone buying car also buying a vacuum)

Two-way Tables: A two-way table is a visual representation of the possible relationships between two sets of categorical data. The categories are labeled at the top and the left side of the table, with the frequency (count) information appearing in the four (or more) interior cells of the table. The "totals" of each row appear at the right, and the "totals" of each column appear at the bottom.

The rows/columns determine the ***condition***. Using the example above, the probability that you select a left-handed person, given that it is a girl is the number of left-handed girls divided by the total number of girls → $P(Left-handed|Girl) = \frac{10}{23} \approx .43$. The ***condition*** in this problem is a ***girl*** therefore, the number of girls represents the total of the conditional probability.

Two-Way Relative Frequency Table

	Girls	Boys	Totals
Left-handed	$\frac{10}{23} \approx .43$	$\frac{15}{27} \approx .56$	$\frac{25}{50} = .50$
Right-handed	$\frac{12}{23} \approx .52$	$\frac{8}{27} \approx .30$	$\frac{20}{50} = .40$
Ambidextrous	$\frac{1}{23} \approx .04$	$\frac{4}{27} \approx .15$	$\frac{5}{50} = .10$
Totals	$\frac{23}{23} = 1.00$	$\frac{27}{27} = 1.00$	$\frac{50}{50} = 1.00$

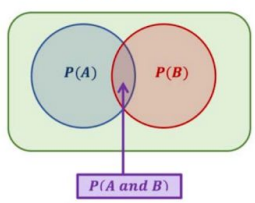
Conditional relative frequencies

Independence: when it is said that two events are independent of each other, it means that the probability of one event occurring does not affect the probability of the other event occurring. Think about it: you rolling a 5 on a dice does not affect you flipping a coin and getting heads.

Joint Probability: the likelihood that two events will occur together.

The Venn diagram shows that when you include everything in both sets the middle region is included twice, therefore you must subtract the intersection region out once. The probability for calculating **joint** events is…

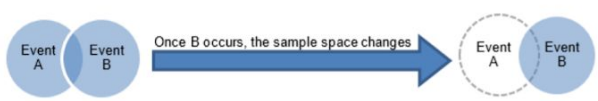
$$P\ (A\ or\ B) = P(A) + P(B) - P\ (A\ and\ B)$$



Students may recognize that if two events A and B are mutually exclusive, also called **disjoint,** the rule can be simplified to $P\ (A\ or\ B) = P\ (A)\ +\ P\ (B)$ since for mutually exclusive events $P\ (A\ and\ B) = 0$.

Marginal Probability: this is the probability of an event without regard to the outcome of another variable.

The sample space of an experiment can be modeled with a Venn diagram such as:



So, the $P(A|B) = \frac{P(A\ and\ B)}{P(B)}$

(**Note**: Two events must be independent to apply the general multiplication rule.)

$$P(A \text{ and } B) = P(A)P(B|A) = P(B)P(A|B)$$

The general rule can be explained based on the definitions of independence and dependence. Events are either independent or dependent.

- Two events are said to be **independent** if the occurrence of one event does not affect the probability of the occurrence of the other event.
- Two events are **dependent** if the occurrence of one event does, in fact, affect the probability of the occurrence of the other event.

Practice problems

1. A number cube is rolled 24 times and lands on 2 four times and on 6 three times.

a. Find the experimental probability of landing on a 2.

b. Find the experimental probability of *not* landing on a 6.

c. Compare the experimental probability you found in part a to its theoretical probability.

d. Compare the experimental probability you found in part b to its theoretical probability.

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